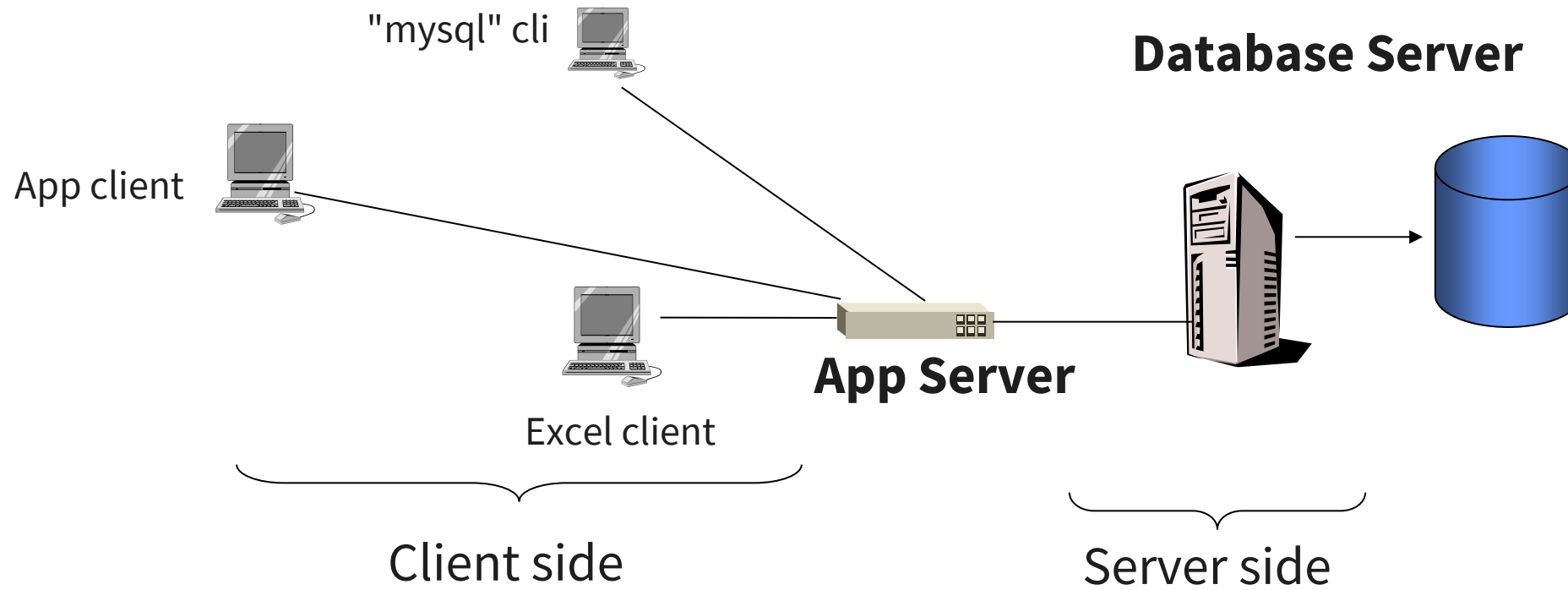


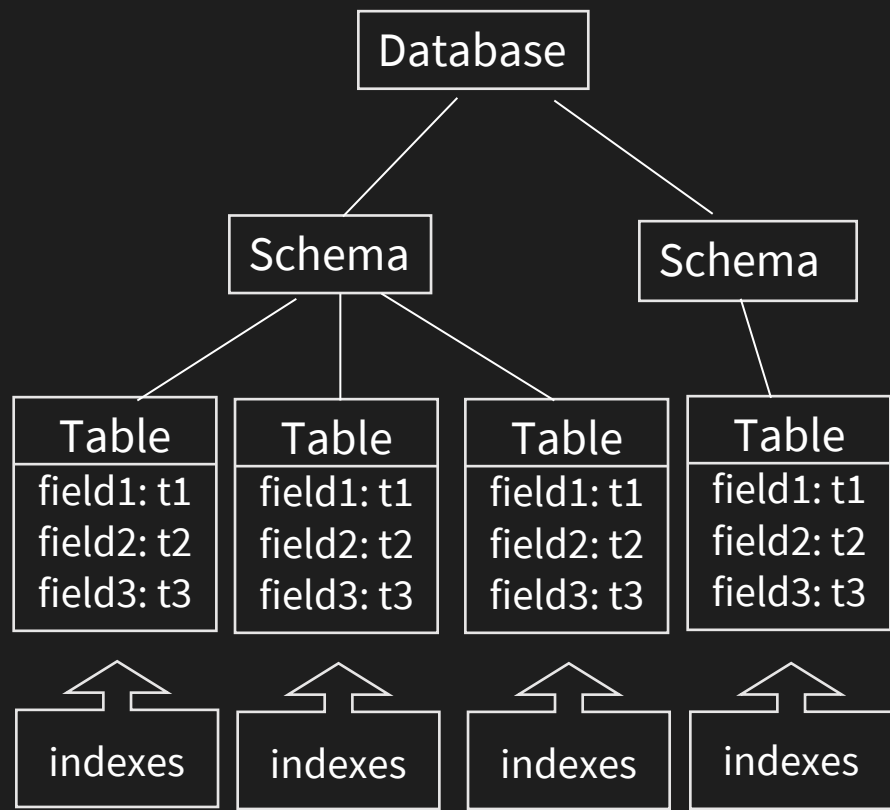


SQL

SHENKAR

BOAZ SCHNEIDER





מבנה בסיס נתונים

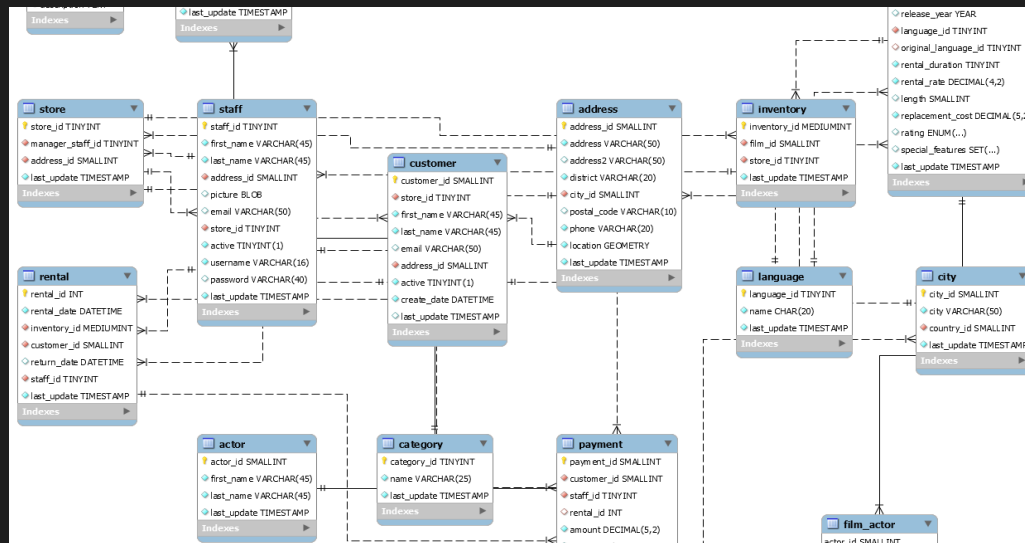
בסיס נתונים מכיל סכימה,
המתארת את מבנהו.

סכימה יכולה להכיל:

tables - containing data

index files - for fast lookup of
data

stored procedures, constraints,
triggers, and more



Every column has:

- a name
- a data type and length

To view the structure of a table use:

`DESCRIBE tablename`

describe city;						
Result Grid Filter Rows: Export: Wrap Cell Content:						
	Field	Type	Null	Key	Default	Extra
▶	ID	int	NO	PRI	HULL	auto_increment
	Name	char(35)	NO			
	CountryCode	char(3)	NO	MUL		
	District	char(20)	NO			
	Population	int	NO		0	

SHOW columns FROM tablename

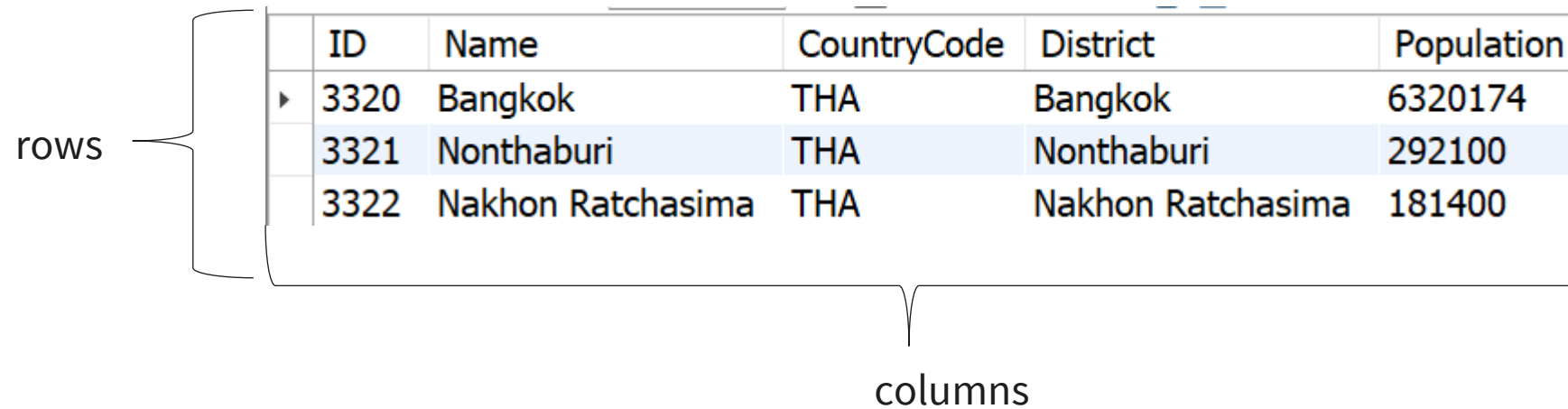
shows the same information as DESCRIBE *tablename*.

SHOW columns FROM City;						
Result Grid Filter Rows: Export: Wrap Cell Content:						
	Field	Type	Null	Key	Default	Extra
▶	ID	int	NO	PRI	HULL	auto_increment
	Name	char(35)	NO			
	CountryCode	char(3)	NO	MUL		
	District	char(20)	NO			
	Population	int	NO		0	

Columns may have a default value to use if a value is not assigned explicitly.

תוכן טבלה

- טבלה מחזיקה את המידע בצורה של רשומות (**rows\ records**).
- רשומה היא צרוף של העמודות (**columns\ fields**).
- כל רשומה מחזיקה צרוף אחד של נתונים.



The diagram shows a table with 6 columns and 4 rows. The first row is the header. The next three rows are data rows. A bracket on the left side of the table, spanning the three data rows, is labeled 'rows'. A bracket at the bottom of the table, spanning all six columns, is labeled 'columns'.

ID	Name	CountryCode	District	Population
3320	Bangkok	THA	Bangkok	6320174
3321	Nonthaburi	THA	Nonthaburi	292100
3322	Nakhon Ratchasima	THA	Nakhon Ratchasima	181400

שדה מפתח

- לכל טבלה יש שדה מפתח (primary key) - המייצג באופן יחודי רשומה.
- שדה המפתח משמש גם לקשר עם טבלאות אחרות (FK).

	ID	Name	CountryCode	District	Population
▶	3320	Bangkok	THA	Bangkok	6320174
	3321	Nonthaburi	THA	Nonthaburi	292100
	3322	Nakhon Ratchasima	THA	Nakhon Ratchasima	181400

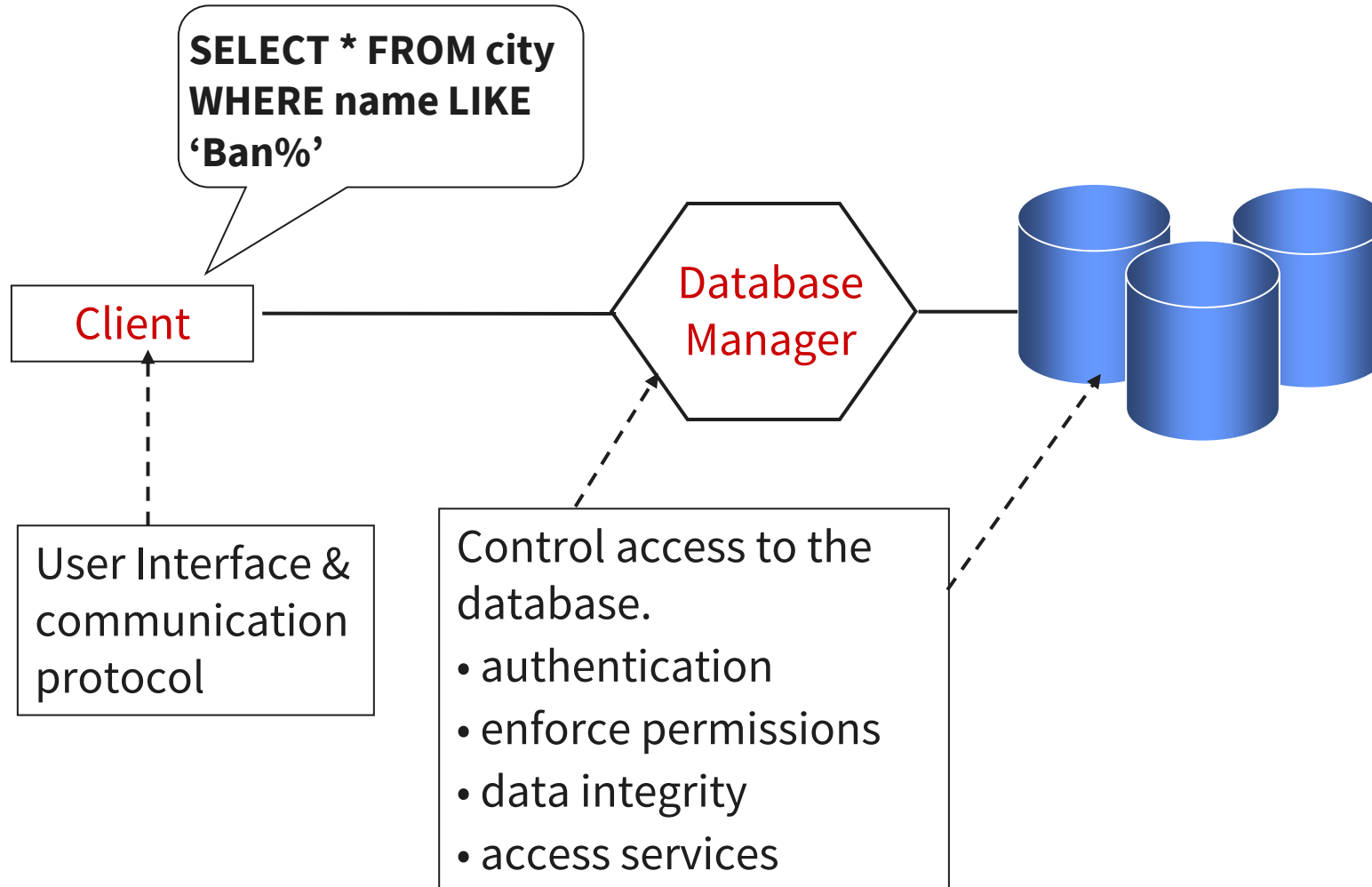
ID is the primary key in City table.

SQL – STRUCTURE QUERY LANGUAGE

- DDL - Data Definition Language - Commands that define a database constraints
 - Create database
 - Create, Drop, Alter tables
- DML - Data Manipulation Language - Commands that maintain and query a database
 - Pull data.
 - Insert new data
 - Update
 - Delete
- DCL - Data Control Language - Commands that control a database, including administering privileges and committing data

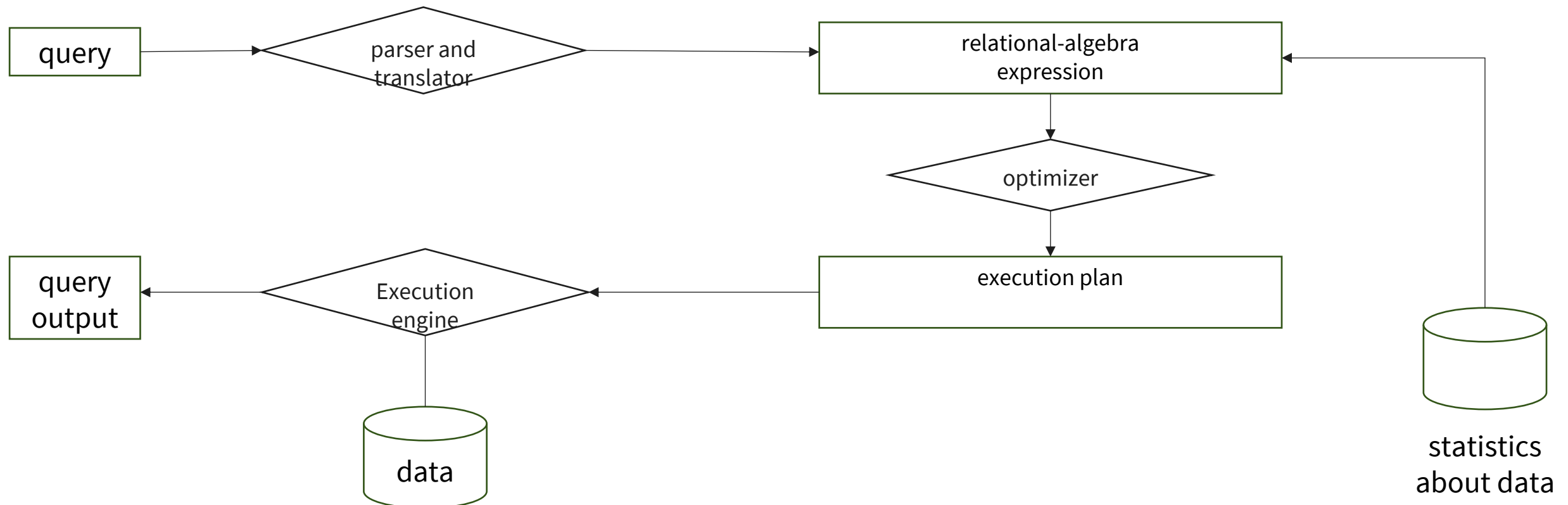
ארבעת הפעולות הבסיסיות בבסיס נתונים

SELECT	query (search) the data
INSERT	add new records to a table(s)
UPDATE	modify existing record(s)
DELETE	delete record(s) from a table



תהליך עיבוד שאילתה:

1. ניתוח ועיבוד
2. אופטימיזציה
3. הרצה

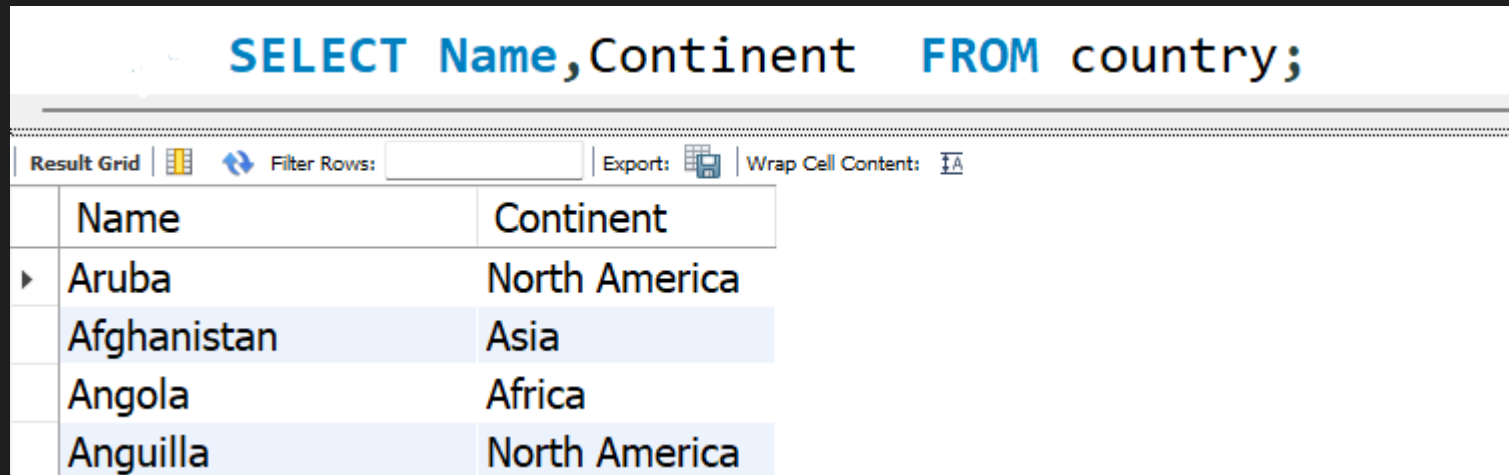


תישאול מידע מטבלה:

SELECT displays field values from a table:

SELECT field1, field2, field3 FROM table ;

- displays ALL rows from the table.
- use LIMIT *number* to limit how many results.



The screenshot shows a database query interface. At the top, the SQL query `SELECT Name, Continent FROM country;` is entered. Below the query, there is a toolbar with options like "Result Grid", "Filter Rows", "Export", and "Wrap Cell Content". The results are displayed in a table with two columns: "Name" and "Continent". The table contains the following data:

Name	Continent
Aruba	North America
Afghanistan	Asia
Angola	Africa
Anguilla	North America

SELECT STATEMENT WITH * (ALL)

- Display values for **all fields** in table:

SELECT * FROM tablename ;

SELECT * FROM country;

Result Grid  Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 									
Code	Name	Continent	Region	SurfaceArea	IndepYear	Population	LifeExpectancy	GNP	
ATA	Antarctica	Antarctica	Antarctica	13120000.00	NULL	0	NULL	0.00	
ATF	French Southern ter...	Antarctica	Antarctica	7780.00	NULL	0	NULL	0.00	
BVT	Bouvet Island	Antarctica	Antarctica	59.00	NULL	0	NULL	0.00	

שאלתה עם תנאים:

- Select columns from a table that match some criteria:

```
SELECT field1, field2, field3  
FROM table  
WHERE condition  
ORDER BY field1,... [ASC|DESC];
```

- Example: cities with population > 5 M

```
SELECT * FROM City  
  WHERE population > 5000000  
  ORDER BY population DESC;
```

WHERE CONDITIONS

name = 'Bangkok'

name LIKE 'Bang%'

pattern match

population >= 100000

population < 500000

gnp <> 0

<> not equals

grade IN ('A','B','C','D','F')

contained in set

שימוש במחרוזות:

- Use **single quote mark** around String constants.

```
SELECT * FROM Country  
WHERE name = 'Thailand';
```

```
SELECT * FROM City  
WHERE Name = 'Los Angeles';
```

MORE SELECT

- `select * from actor;` -- select all columns from table
- `select first_name, last_name from actor;` -- select specific columns by comma
- `select first_name fname from actor;` -- alias
- `select distinct first_name from actor;` -- remove duplicate
- `select distinct first_name from actor order by first_name;` -- order result set by name

MORE WHERE

- `select * from customer where store_id = 1;`
- `select * from customer where store_id = 1 and first_name = 'LINDA';`
- `select * from customer where store_id = 1 and first_name like 'mar%';`
- `select address_id,first_name,last_name from customer where address_id = 5 or address_id = 11;`
- `select address_id,first_name,last_name from customer where address_id in (5,11);`

תירגול

- Return all names of all the languages (sorted Z-A)?
- Return all addresses from the city 'Lethbridge'.
- Return the full names (first and last) of all the actors and costumers whose first name is the same as the first name of the actor with ID 8.